CIE AS PHYSICS (9702) • QUESTION PAPER

1.4 Scalars & Vectors

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Structured Questions (Paper 2) **Total:** 6 Questions (60 Marks)

Question 1 Easy [16 marks]

Part (a) [2 marks]

Distinguish between scalar and vector quantities.

Part (b) [6 marks]

(i) Energy is a scalar quantity. State **three** other scalar quantities.

[3]

(ii) Force is a vector quantity. State **three** other vector quantities.

[3]

Part (c) [4 marks]

Fig. 1.1 shows a block at rest on a slope.

Three forces, weight *W*, normal contact force *N* and the frictional force *F* act on the block. The forces are in equilibrium.

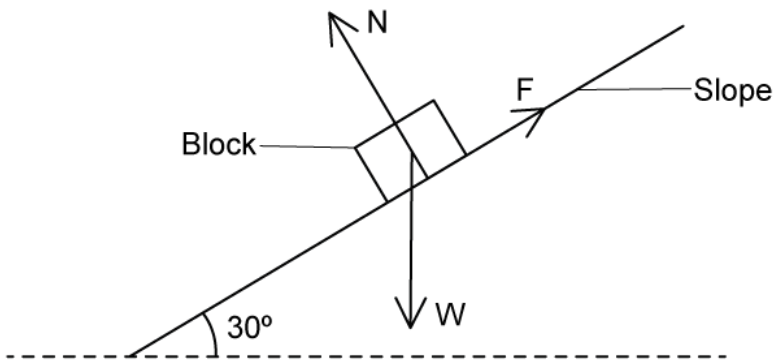


Fig. 1.1 (not to scale)

(i) Explain what is meant by 'the forces are in equilibrium'.

[2]

(ii) Complete the vector triangle in Fig. 1.2. Label the missing vector and the angle between *W* and *N*.

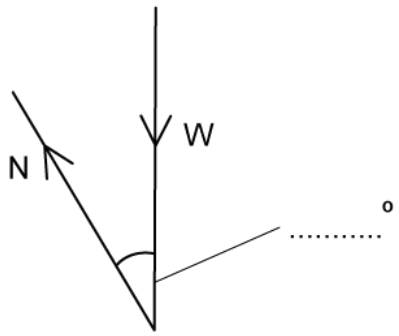


Fig. 1.2

[2]

Part (d) **[4 marks]**

The frictional force has a magnitude of 60 N.

By resolving the vector, calculate the component of the force that acts

(i) horizontally

[2]

(ii) vertically.

[2]

Question 2 Easy [9 marks]

Part (a) [2 marks]

State the definition of a vector quantity.

Part (b) [3 marks]

A list of scalar quantities and their SI unit is given.

Quantity	SI unit
Temperature	
	kg m s^{-1}
Mass	

Complete the information missing from the table.

Part (c) [2 marks]

A person cycles along a path every day, from their front garden at X to a bench by the nearby lake at Y, as shown in Fig. 1.1.

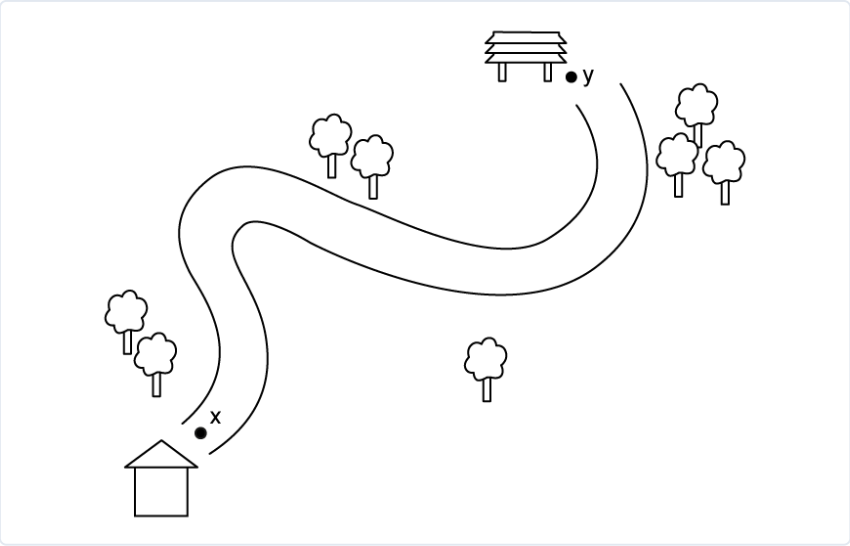


Fig. 1.1

(i) Sketch a line on the image that would indicate the distance travelled by the person between X and Y. Label this distance *d*.

[1]

(ii) State whether the distance travelled *d* by the person is greater than, equal to, or less than their displacement between X and Y.

[1]

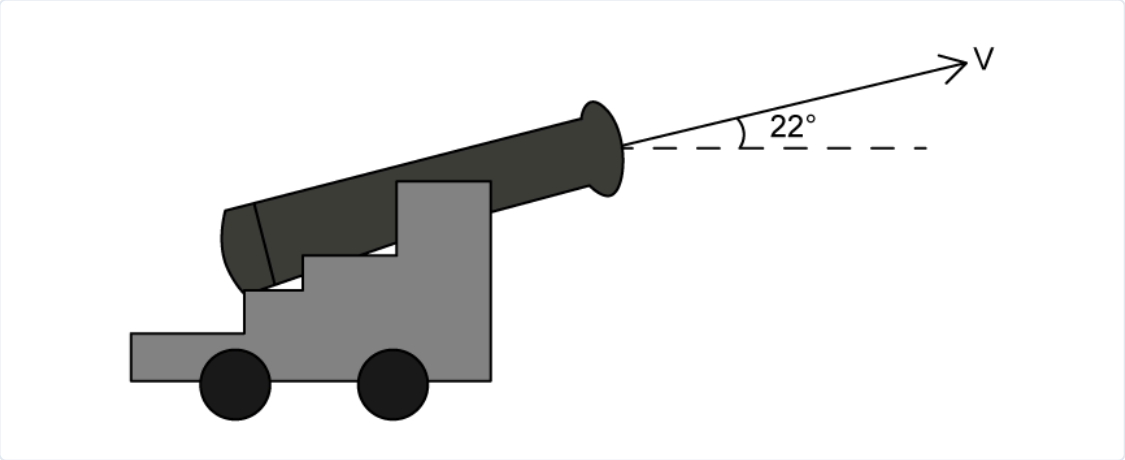
Part (d) [2 marks]

The person continues their journey and walks along the same path back home, from Y to X. The total distance for their complete journey which was logged by their smartwatch is 4.3 km. State and explain what the total displacement for their journey is.

Question 3 Easy [7 marks]

Part (a) [2 marks]

A small cannon fires a projectile at an angle of 22° to the horizontal with an initial velocity *v*.



Calculate the vertical component of velocity if its initial velocity has a magnitude of v = 15 m s^{-1}.

Part (b) [1 mark]

State the direction of the horizontal component of velocity.

Part (c) [2 marks]

The cannon fires a second projectile. The initial horizontal component of velocity is 8.4 m s^{-1} and the initial vertical component of velocity is 5.8 m s^{-1}.

Calculate the initial velocity, v_2, of the second projectile.

Part (d) **[2 marks]**

Calculate the angle to the horizontal that the second projectile was fired at.

Question 4 Medium [9 marks]

Part (a) [7 marks]

The resistive force F acting on a spherical raindrop falling vertically in the air can be described by the relationship

$$F = krv$$

where r is the radius of the raindrop, v is the velocity of the raindrop, and k is a constant.

Fig. 1.1 shows the forces acting on the raindrop as it accelerates downwards.

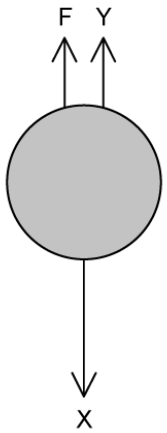


Fig. 1.1

The raindrop has a radius of 1.5 mm and falls at a velocity of 3.5 m s^{-1}. The constant *k* has a magnitude of 0.0124 in air.

Other than the resistive force *F*, two other forces, X and Y, act on the raindrop drop as it falls through the air.

(i) State the names of forces X and Y.

[2]

(ii) Calculate the magnitude of the resistive force *F* in μN.

[2]

(iii) Force X has a value of 137 μN and force Y has a value of 0.18 μN.

Using Fig. 1.1, determine the magnitude and direction of the resultant force on the raindrop.

[3]

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Part (b) [2 marks]

On a windy day, the resultant force on the raindrop is 105 μN at an angle of θ to the vertical as shown in Fig. 1.2.

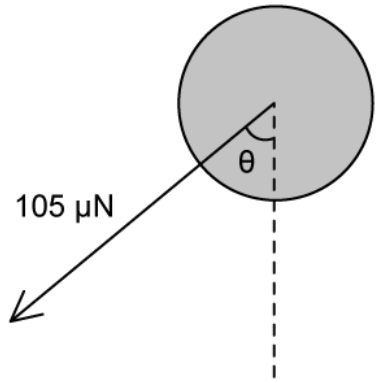


Fig. 1.2

Use your answer to **(a)iii** to show that the angle θ between the resultant and the vertical is about 47°.

Question 5 Medium [6 marks]

Part (a) [2 marks]

(i) State the difference between a scalar quantity and a vector quantity.

[1]

(ii) Underline all the vectors in the list below.

power	length	density	velocity	momentum

[1]

Part (b) [4 marks]

Two gusts of wind exert forces of magnitude 5.5 N and 11.9 N on a wind turbine at a point O. Both forces act away from point O and the angle between them is 30°.

Fig. 1.1 shows two lines at an angle of 30° to one another.

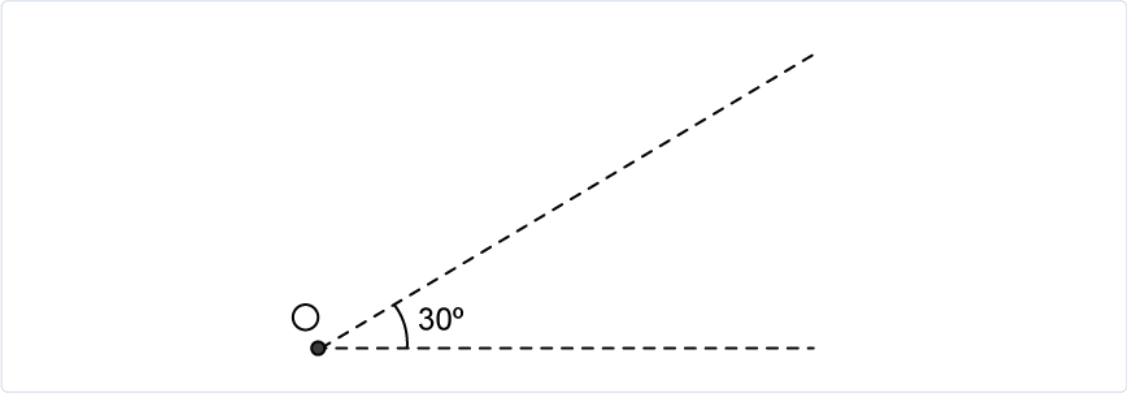


Fig. 1.1

On Fig. 1.1, draw a vector diagram to determine the magnitude of the resultant of the two forces.

Question 6 Medium [13 marks]

Part (a) [4 marks]

Fig. 1.1 shows a 2 tonne wrecking ball being used on a demolition site.

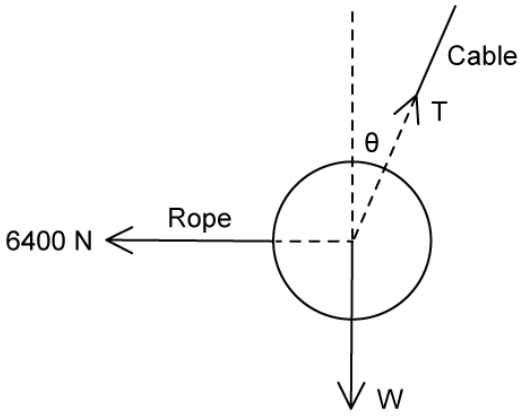


Fig. 1.1

The ball is suspended from a cable which makes an angle θ with the vertical. The ball is pulled into the position shown by a rope that is kept horizontal. The tension in the rope is 6400 N.

Write expressions for tension T using

(i) the vertical component of the tension in the cable

[2]

(ii) the horizontal component of the tension in the cable.

[2]

Part (b) [4 marks]

Determine

(i) the angle θ the cable makes with the vertical

[2]

(ii) the magnitude of the tension T in the cable

[2]

Part (c) **[3 marks]**

On the same demolition site, there is a uniform beam supported by two light cables, **AB** and **AC**, which are attached to a single steel cable from a crane, as shown in Fig. 1.2.

The beam is stationary and in equilibrium.

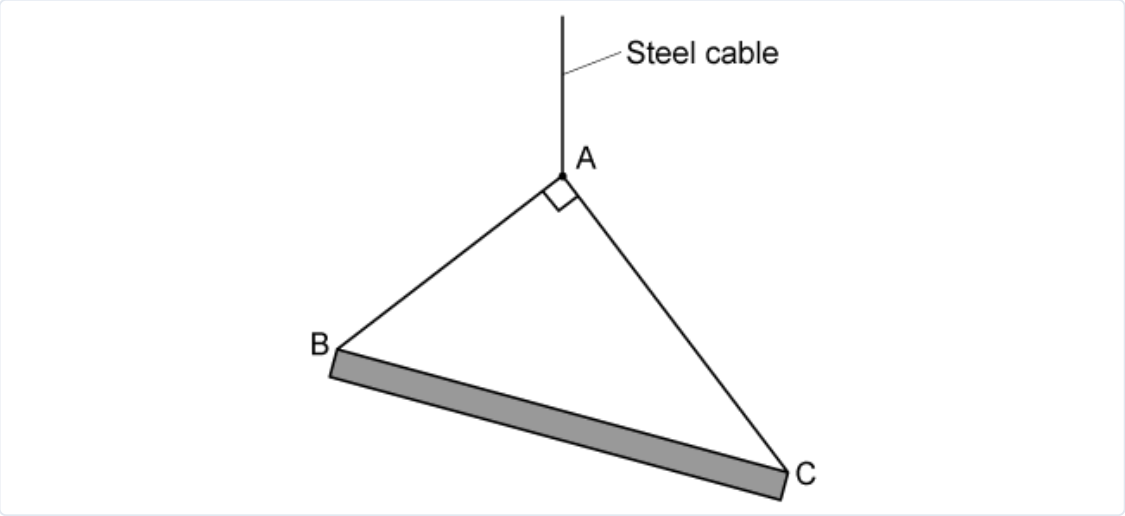


Fig. 1.2

Draw the vector triangle for this situation labelling the tension in both cables and the weight of the beam.

Part (d) **[2 marks]**

The tension in the cable **AB** is 9 N and the tension in the cable **AC** is 12 N.

Calculate the resultant force required in the beam **BC** to keep the system in equilibrium.